

Surya Rao Karri

Core Banking | Full Stack Dev & Devops | [Linkedin](#) | karri.suryarao6@gmail.com | [0642574852](tel:0642574852) |Bangkok, TH.
Portfolio Website: <https://suryaraokarri.github.io/>

Software Engineer with 6+ years of experience developing distributed systems using Java Spring Boot and Microservices. Skilled in Event Driven Architecture (Kafka), cloud deployments with Kubernetes, and building automation tools. Background involves modernizing core banking applications and ensuring system security.

SKILLS

Programming & Frameworks:	Java 17/21, Spring Boot/Cloud, Python, Microservices, JavaScript/TS, RESTful APIs, Event-Driven Architecture, Apache Kafka, Redis.
DevOps & Cloud:	Kubernetes (K8s), Docker, AWS (EC2, S3, RDS, Lambda), Jenkins, GitLab CI, Terraform.
Database:	PostgreSQL, MongoDB (NoSQL), SQL optimization, Transaction Management.
Security & Quality:	OAuth2, JWT, SonarQube, Prometheus, Grafana, Vulnerability Assessment.
AI Tools:	Ollama (local LLMs, Gemma 4), GitHub Copilot, Claude, Gemini.
Languages:	English (Fluent), French (Intermediate).

EXPERIENCE

Yip In Tsoi & Co. Limited.	Aug 2019 – Present
Senior Banking Application Developer	Bangkok, Thailand
- Build Spring Boot APIs and support for Microservices, enabling independent deployments and reducing dependencies for core banking systems. - Event-Driven Design for Anti-Money Laundering (AML) system using Apache Kafka for real-time asynchronous data ingestion. - Migration of monolithic legacy modules into scalable Java Spring Boot REST APIs. - Manage Docker, Kubernetes and automated CI/CD pipelines (Jenkins/GitLab). - Reduced API latency by using Redis caching and SQL tuning, and implementing OAuth2/JWT security. - Managed production stability using Prometheus and Grafana for real-time microservice health monitoring and distributed tracing. - FIS, Profile, MUMPS, PSL, GTM, Batch processing, Go Top up Mobile Banking, Web service transactions.	
Baan Banchi Co., Ltd.	Aug 2018 – Aug 2019
Software Developer Engineer	Bangkok, Thailand
Developed and maintained a WPF-based accounting application, contributing to core features and bug fixes.	

CONSULTANT & FREELANCE

Freelance	Aug 2023 - Present
Automation Tools Developer - Own Projects & Build custom tools for SMEs	Bangkok, Thailand
Developed robust tools for complex, dynamic applications (e.g., maps, web applications, extensions) to perform automatic operations or to collect large information and ingesting and storing approx 4 million tokens into Vector Databases to power business insights and RAG workflows.	

- Created custom Chrome extensions to automate workflows and reduce manual tasks on web platforms.
- Built and deployed a data scraper, currently in use by university students in the USA.

Technologies: Java Spring Boot, Node.JS, Selenium, Google Puppeteer, Natural Language Processing (NLP) and Artificial Intelligence (AI), LangChain, Kafka, Vector DBs.

Equmenopolis, Inc. | Tokyo, Japan (Remote).

Jan 2026 – March 2026

Conversational AI WebRTC Engineer (Consultant - Freelance)

Tokyo, Japan

- Engineered real-time WebRTC pipelines (Agora, LiveKit) to synchronize Unity avatar video/audio streams with live user media.
- Developed Python and FastAPI microservices utilizing Redis Pub/Sub to manage WebRTC signaling and coordinate system states.
- Deployed workloads on RunPod (RTX 3090) and resolved critical CPU bottlenecks by configuring hardware-accelerated Xorg/X11 rendering.
- Profiled GStreamer pipeline performance using GstShark to identify and minimize streaming latency.
- Technologies: Python 3.11, WebRTC (LiveKit, Agora), GStreamer, FastAPI, Redis, Docker, RunPod, PyTorch, ONNX Runtime, GCP, GstShark.

LEET Intelligence Co., Ltd.

Oct 2019 – Dec 2023

SD Team Lead - (Consultant - Freelance)

Pathum Thani, Thailand

- Developed apps for Web, Android, and IOS | GIS applications with LULC, Loss/Gain, and other classes.
- Responsible for code base maintenance, and deployment pipeline.
- End-to-end project development.
- Technologies: NestJS, AWS, GEE API, React with TypeScript.

DOPA Thailand | Smart Device Registration Application - Freelance

Sep 2021 – Dec 2021

- Developed an android application for Biometrics Enrollment Device to register users.
- Capture fingerprint in wsq format from sensors, smart ID card details from NFC sensor using APDU Command and Capture user photo, print the form.
- Windows Application UI using ElectronJS and Nodejs to build API | Tech: Java, Kotlin, Android.

EDUCATION

Asian Institute of Technology

Jan 2017 - Aug 2018

Master of Engineering; Major in Computer Science

Pathum Thani, Thailand

- Research in developing Computer Security models to detect data breaches using Prolog language.

Jawaharlal Nehru Technological University Hyderabad

Sep 2013 - Dec 2016

Bachelor of Technology; Major in Computer Science

Hyderabad, Telangana, India